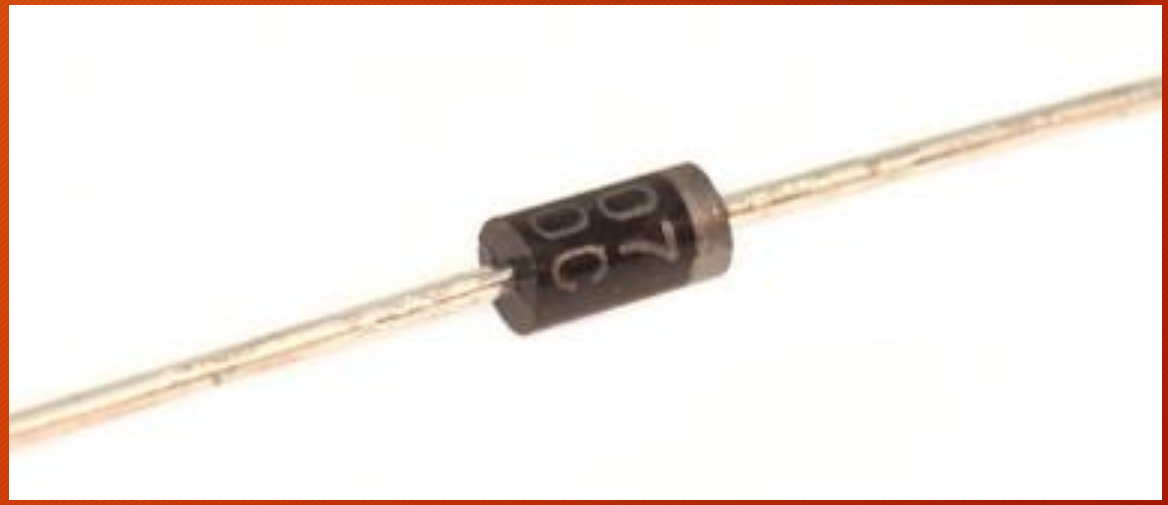
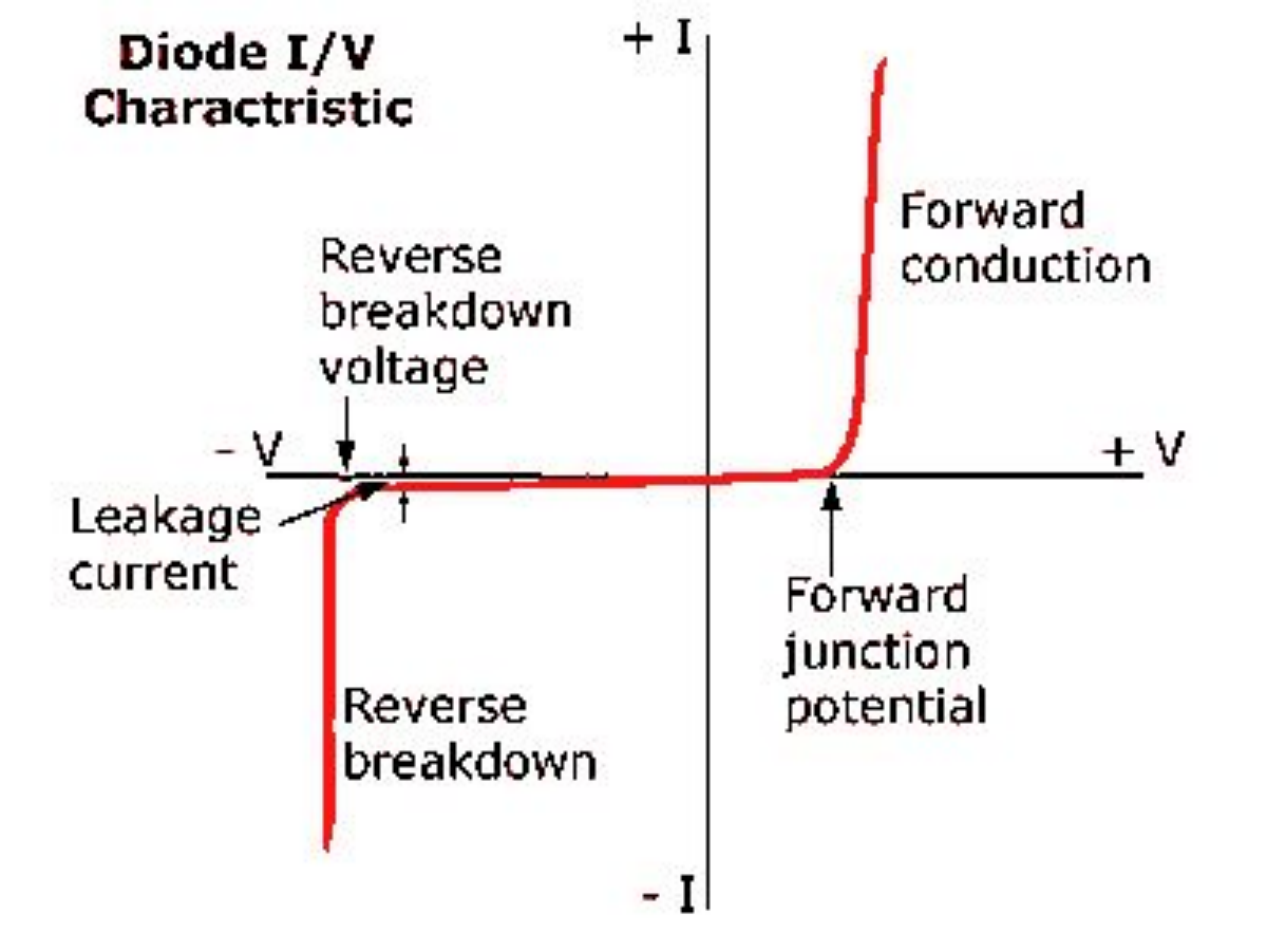
Diode (1N4007)



I/V Characteristics of a Diode



List of Parameters	SILION DIODE (1N4007)
Maximum Forward Current	1A
Maximum Reverse Current	5.0μA
Maximum Forward Voltage	0.8V
Maximum Reverse Voltage	1000V
Maximum Power Dissipation	30mW

Transistor -7809



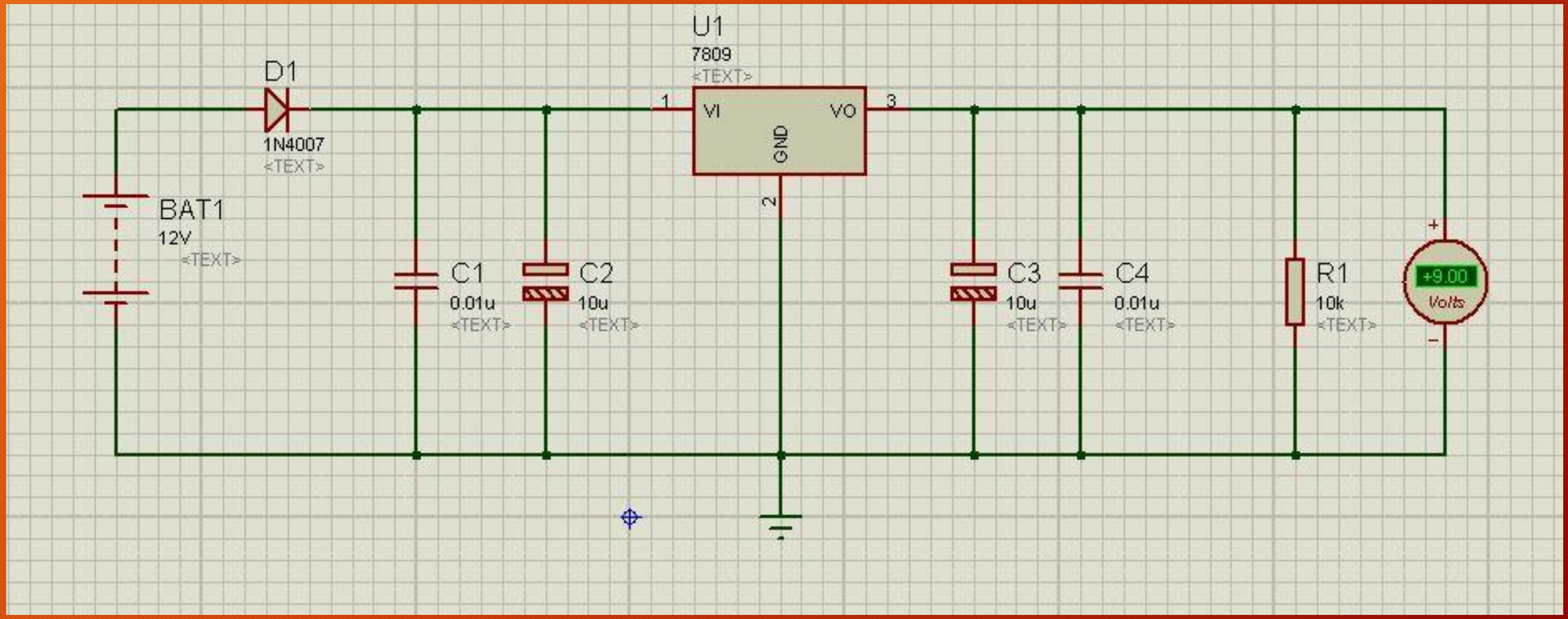
78xx Family

- The **78xx** (sometimes **L78xx**, **LM78xx**, **MC78xx**...) is a family of self-contained fixed linear voltage regulator integrated circuits.
- For ICs within the family, the *xx* is replaced with two digits, indicating the output voltage.
- The 78xx line are positive voltage regulators: they produce a voltage that is positive relative to a common ground. There is a related line of **79xx** devices which are complementary negative voltage regulators.

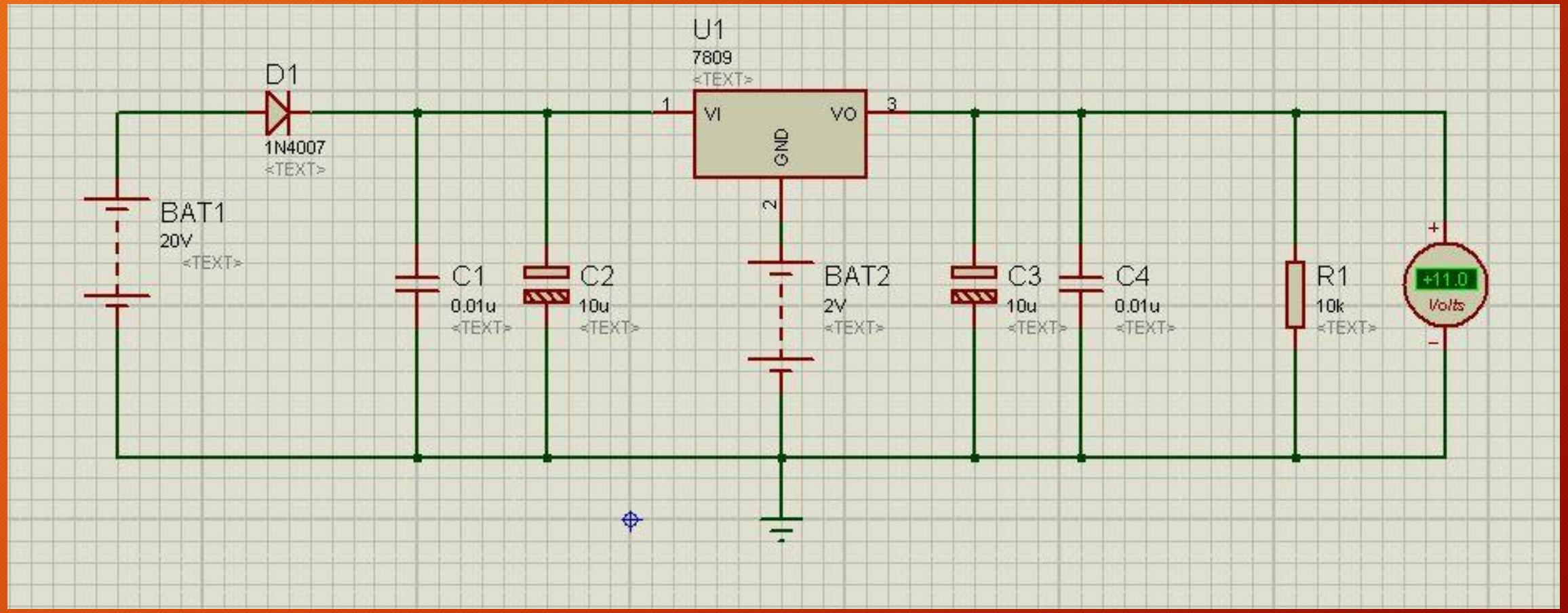
7809 Characteristics

- Minimum Input Voltage: 11V
- Output Voltage: 9V (w.r.t Ground)
- Minimum Voltage Drop: 2V

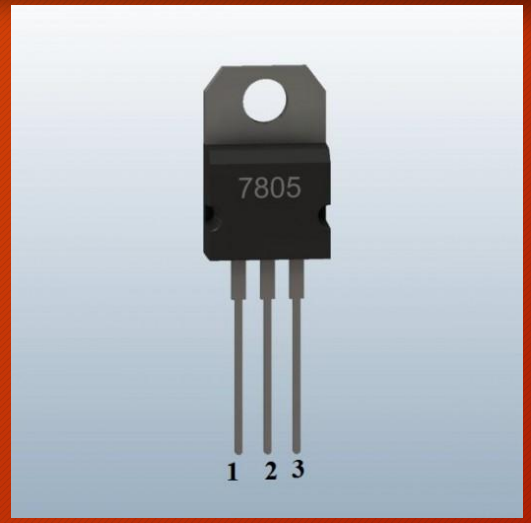
7809 Transistor Schematic



Can we create any voltage from 7809?



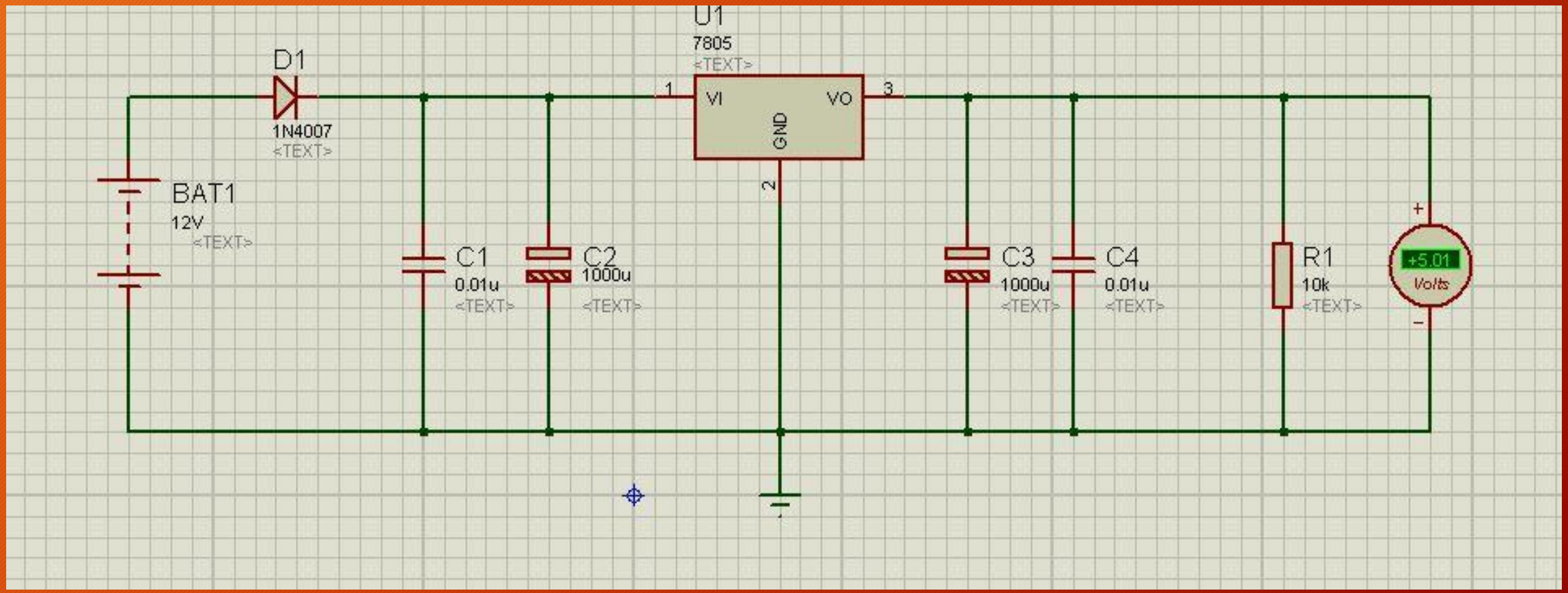
Transistor -7805



7809 Characteristics

- Minimum Input Voltage: 7.3V
- Output Voltage: 5V (w.r.t Ground)
- Minimum Voltage Drop: 2V

7809 Transistor Schematic



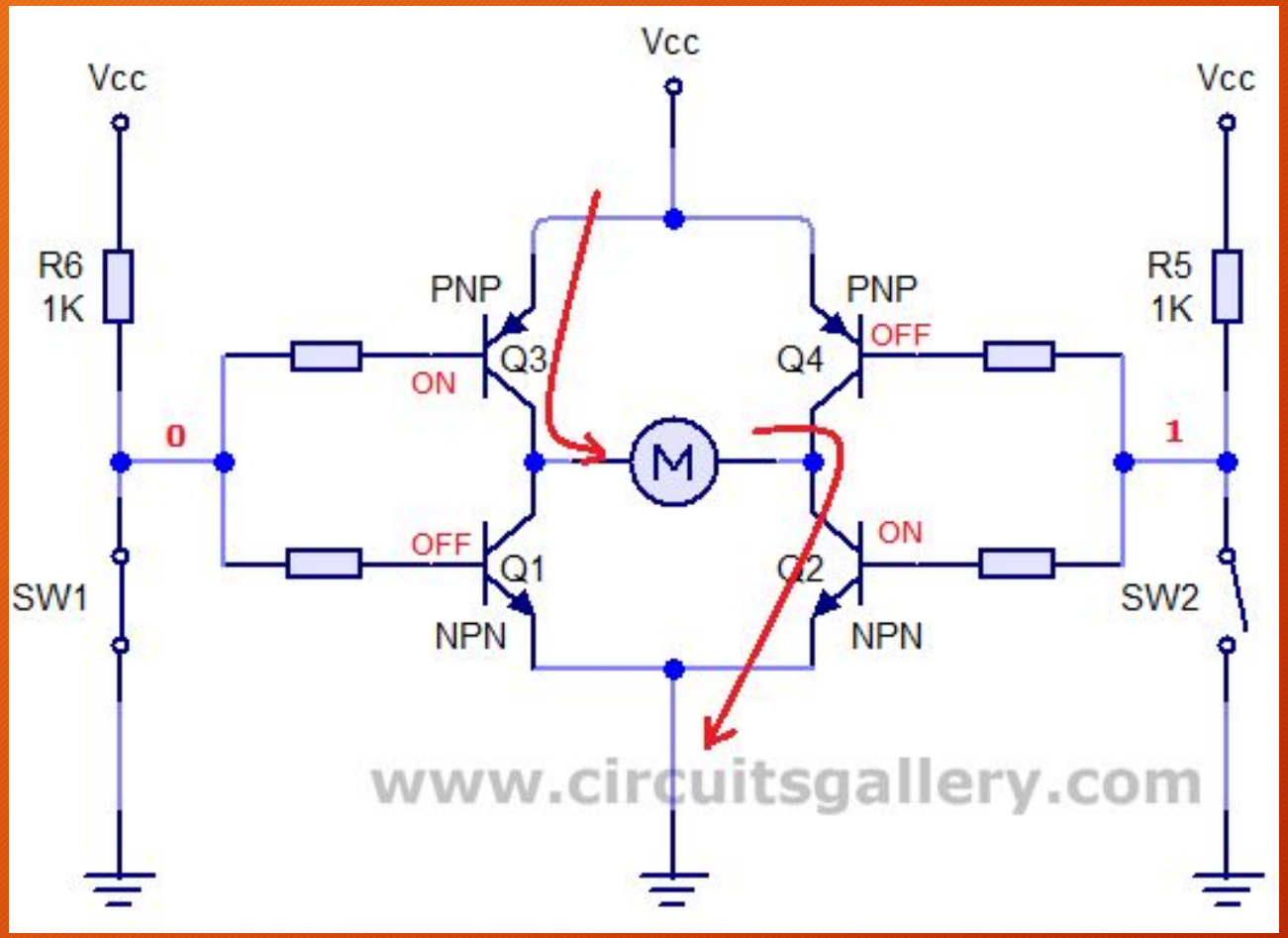
Motor Driver Board (L293D)

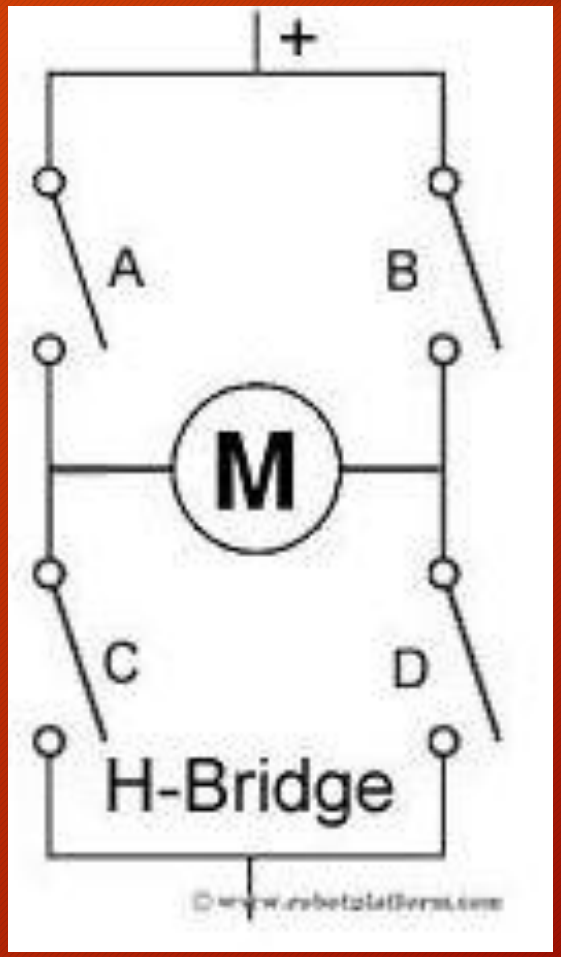


Why do we use Motor Driver Board ?

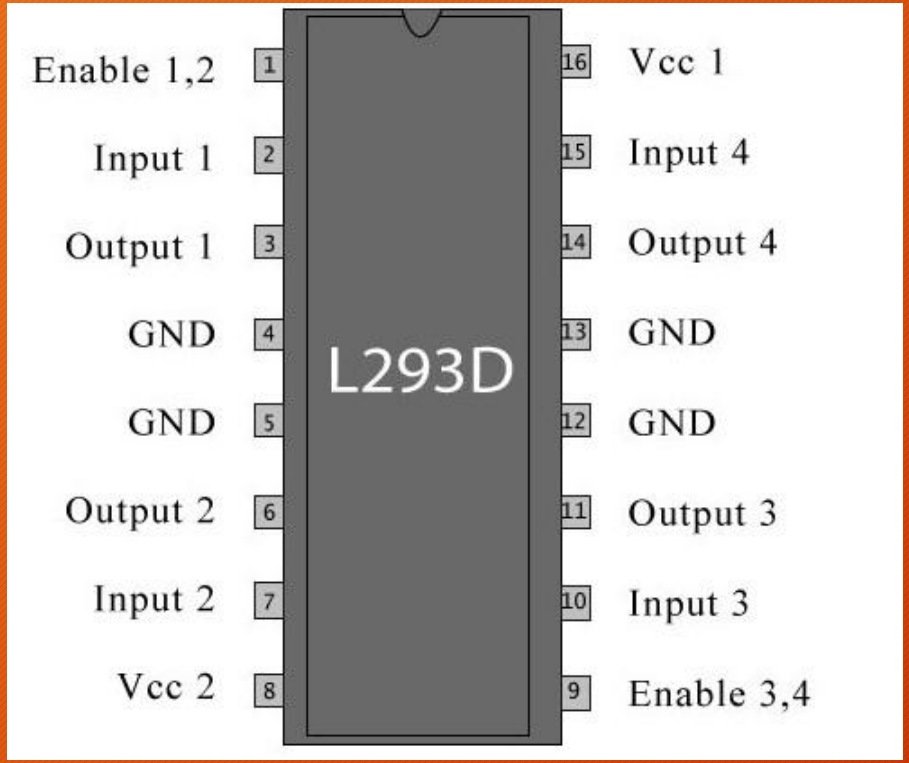
- Maximum Current from Arduino: 50mA
- Current Requirement of Motor: 600mA
- Motor driver is basically a current amplifier which takes a low-current signal from the microcontroller and gives out a proportionally higher current signal which can control and drive a motor. In most cases, a transistor can act as a switch and perform this task which drives the motor in a single direction.

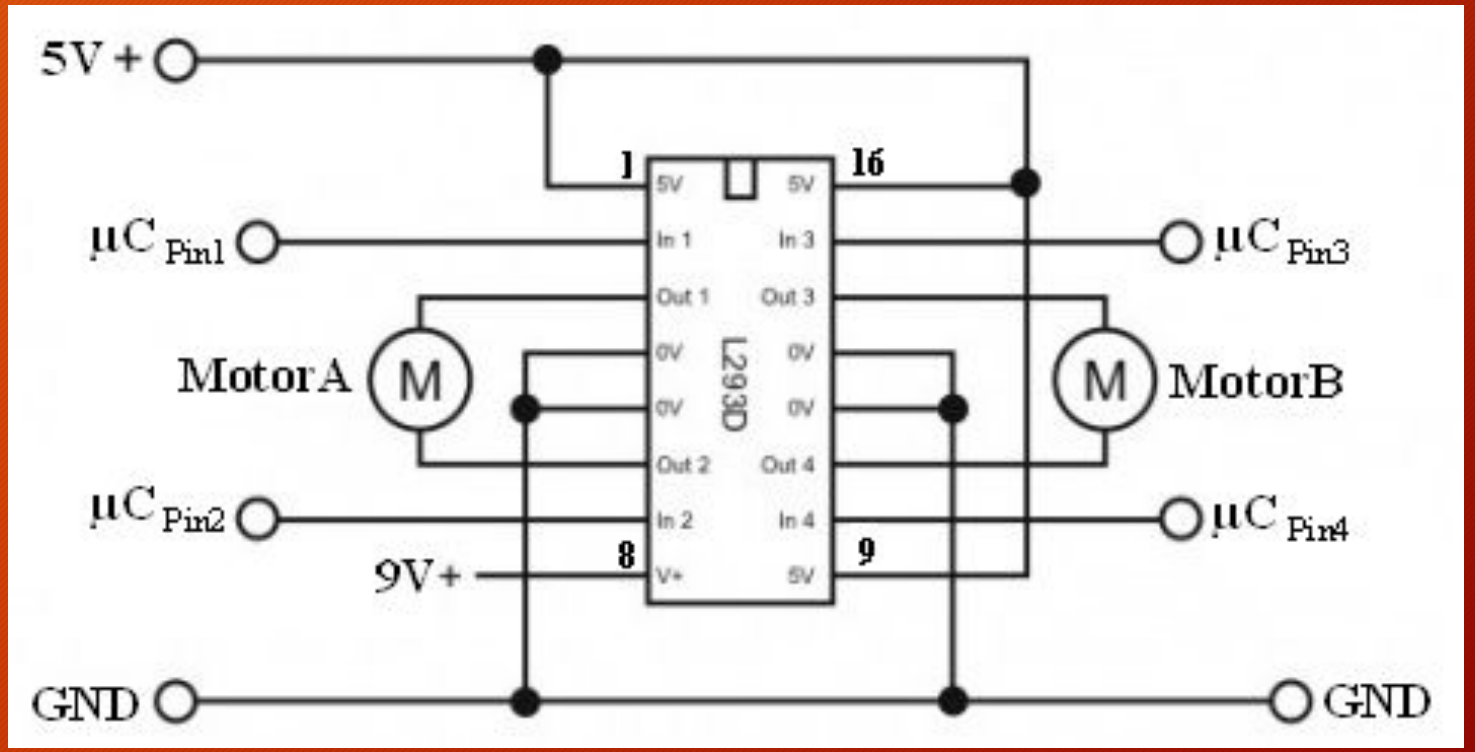
H-Bridge


Vcc
Vcc
Vcc
R6
1K
R5
1K
PNP
PNP
Q3
Q4
ON
OFF
0
1
M
OFF
ON
Q1
Q2
NPN
NPN
SW1
SW2
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+
A
B
M
C
D
H-Bridge

L293D IC


Enable 1,2
1
Input 1
2
Output 1
3
GND
4
GND
5
Output 2
6
Input 2
7
Vcc 2
8
L293D
16
Vcc 1
15
Input 4
14
Output 4
13
GND
12
GND
11
Output 3
10
Input 3
9
Enable 3,4


5V +
µC Pin1
MotorA
M
9V +
GND
1
16
8
9
5V
In 1
Out 1
0V
0V
Out 2
In 2
V+
In 3
Out 3
0V
0V
Out 4
In 4
5V
L293D
M
MotorB
µC Pin3
µC Pin4
GND